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Introduction

In this report, we look to optimise the performance of a hedge fund's functions for calculation of European call and put option prices. The company currently has two functions to compute option prices, one for call and one for put, that use the binomial model with the Cox-Ross-Rubenstein (CRR) approximation. We look to combine this into one function, improve its computational complexity, and compare different methodologies of pricing to determine which offers the greatest accuracy and lowest computational time. The hedge fund currently offers a 20 month option on an underlying stock, for which the current stock price is £285, the option strike price is £290, the volatility of the stock is 12% and the current interest rate is 2.1%. Our conclusions are based on these parameters over a range of time steps in the binomial model.

The Binomial Model

In this section, we implement the pricing of a vanilla European option on a recombining binomial tree with the approximate CRR method. The unoptimised original algorithm uses a matrix to store values, we look to optimise this. In building our optimised function we choose to use an approach that uses no arrays by utilising the assumed binomial distribution of stock prices, as seen in **Algorithm 1**.

Algorithm 1 Compute the price of a European call option using the approximate binomial CRR model with no arrays

Input: S_0, r, σ, T, K, M , Option Type

Compute $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and $p^* = \frac{e^{r\Delta t} - d}{u - d}$ where $\Delta t = T/M$
if $p^* \leq 0$ or $p^* \geq 1$ **then** Error: Not arbitrage-free **end if**

Initialize $V = 0$, $E[S] = 0$, $E[S^2] = 0$, binomial coefficient $C = 1$

for $j = 0$ **to** M **do**

 Compute $S_{j,M} = S_0 \cdot u^j \cdot d^{M-j}$

 Compute $p_{j,M} = C \cdot (p^*)^j \cdot (1 - p^*)^{M-j}$

 Update $E[S] = E[S] + p_{j,M} \cdot S_{j,M}$

 Update $E[S^2] = E[S^2] + p_{j,M} \cdot S_{j,M}^2$

if call option **then** compute $h(S_{j,M}) = \max(S_{j,M} - K, 0)$

else compute $h(S_{j,M}) = \max(K - S_{j,M}, 0)$

 Update $V = V + p_{j,M} \cdot h(S_{j,M})$

if $j < M$ **then** update binomial coefficient $C = C \cdot \frac{M-j}{j+1}$

end for

Compute $V = e^{-rT} \cdot V$

Output: V , the price of the European option, $E[S]$, the mean stock price, $E[S^2]$, the second moment of the stock price

A potential downside to the use of this algorithm is that it can only be used for the pricing of path independent European options. This is irrelevant in our case, as our company is only concerned with these option types. The main benefits of using this algorithm are the improved time and memory complexity. Table 1 highlights the differences in complexities depending on the use of matrices or arrays. From this it is clear that our approach, using no arrays is superior in terms of complexity. The lower time complexity of our approach can sometimes be misleading due to the potential increase in time complexity to $\mathcal{O}(M^2)$ due to calculations of $M!$. We do not face this issue as we avoid direct calculation of $M!$ and use an iterative approach for calculation of the binomial coefficients.

Function Type	Time Complexity	Memory Complexity
Original Matrix Function	$\mathcal{O}(M^2)$	$\mathcal{O}(M^2)$
Two Arrays Function	$\mathcal{O}(M^2)$	$\mathcal{O}(M)$
One Array Function	$\mathcal{O}(M^2)$	$\mathcal{O}(M)$
No Arrays Function	$\mathcal{O}(M)$	$\mathcal{O}(1)$

Table 1: Time and Memory Complexity of Different Functions

We support this theoretical ideology further by testing the true computational time of our optimised algorithm in comparison to the original algorithm that uses a matrix. We use the given parameters ($S_0 = \text{£}285, r = 2.1\%, \sigma = 12\%, T = 20, K = \text{£}290$) and see how computational time changes as M is increased for both a call and put option. As seen in figure 1, our optimised algorithm is significantly faster than the original version, with the benefits increasing further as M increases. Thus, the practical application supports the theory.

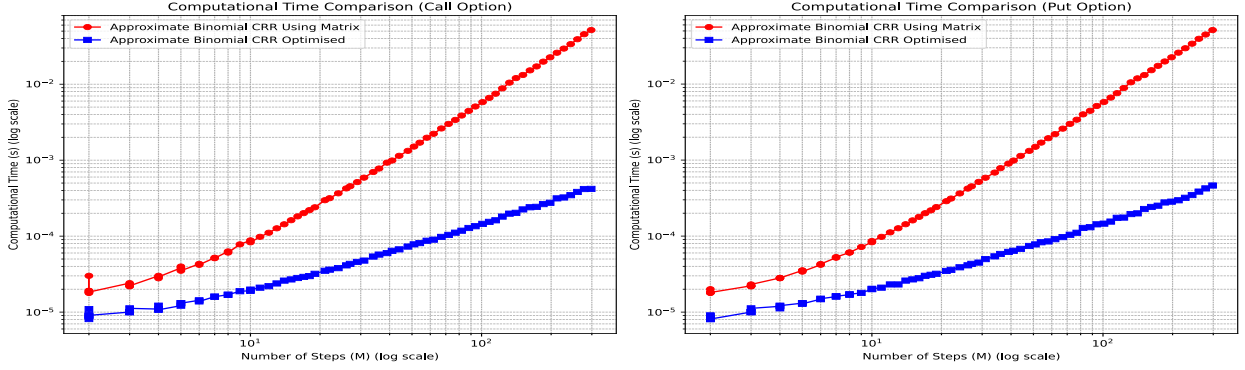


Figure 1: Comparison of Computational Time for the Original Matrix Function and the Optimised (No Arrays) Function

Verification

In this section, we verify the accuracy of our implementation of the algorithm discussed in the previous section by comparison of the price calculated from our function to the true price (using the Black-Scholes model). The equations implemented for calculation of option prices using the Black-Scholes model can be seen below.

$$\text{price of call} = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (1)$$

$$\text{price of put} = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1), \quad (2)$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad (3)$$

$$d_2 = d_1 - \sigma\sqrt{T}. \quad (4)$$

Calculation of these option prices require evaluation of the Φ function. We do this using the `norm.cdf(x)` function from SciPy. It computes:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt. \quad (5)$$

Since this integral has no closed-form solution, SciPy evaluates it using the error function:

$$\Phi(x) = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right) \quad \text{where} \quad \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt. \quad (6)$$

SciPy efficiently computes $\operatorname{erf}(x)$ using numerical approximations such as rational approximations or Chebyshev polynomials, ensuring high precision across different values of x .

We start by an initial comparison of the price for both a call and put option with our function and from the Black-Scholes model under the given parameters ($S_0 = \text{£}285, r = 2.1\%, \sigma = 12\%, T = 20, K = \text{£}290, M = 50$). From table 2, we can see the approximate binomial CRR method returns a price close, although not quite exact for $M = 50$, to the true price for both call and put options.

Model	Call Option Price	Put Option Price
Black-Scholes	20.0425	15.0680
Approximate Binomial CRR	20.1157	15.1413

Table 2: Comparison of Call and Put Option Prices for Black-Scholes and Approximate Binomial CRR

To further convince ourselves of the correctness of our program we check for the convergence of the option price to the Black-Scholes price as M is increased. The theorem states that when the number of time periods (M) goes to infinity the

option price computed by a calibrated binomial model converges to the Black-Scholes price. From theory, we also know the convergence of the approximate binomial CRR model is expected to be irregular. From figure 2, it is clear that our function falls in line with the theorem, so we can be convinced of its correctness.

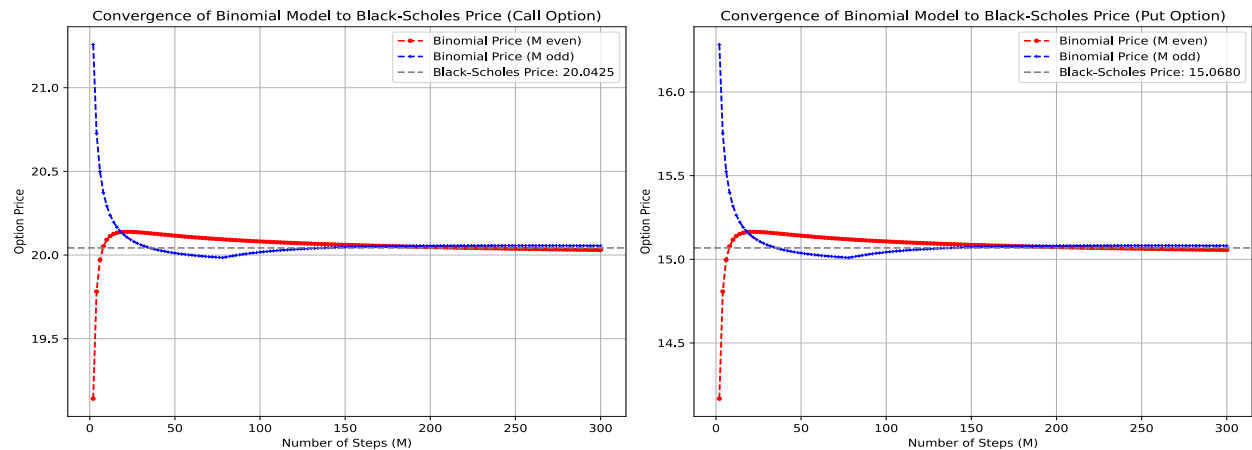


Figure 2: Convergence of Approximate Binomial CRR to Black-Scholes Price

Investigating the Choice of Upward Stock Price Movement

In this section, we start by comparing the mean and the second moment of the stock price computed by the approximate binomial CRR model to the true value (from the Black-Scholes model). From table 3, we can see the mean stock price is the same to 4 decimal places and the second moment of the stock price is fairly close to the true stock price with an absolute error of 4.1096. This is expected, as moment matching is used to choose u and d . Risk-neutral pricing ensures the first moment is correct regardless of M . The slight discrepancy in the second moment is caused by M not being sufficiently large, so there is still discretisation error.

Model	Mean Stock Price	Second Moment of Stock Price
Black-Scholes	295.1516	89230.5154
Approximate Binomial CRR	295.1516	89226.4058

Table 3: Comparison of Mean Stock Price and Second Moment of Stock Price for the Approximate Binomial CRR Model to the True Values

We go onto investigate the stock price distribution of the approximate binomial CRR model compared to the true stock price distribution (from the Black-Scholes model). Looking at the PDF comparison, seen in figure 3, it is clear they follow a similar shape, with the same position of the peak (explaining the shared mean), but the distribution of stock prices from the Black-Scholes model is much smoother. This difference can be explained by the discrete nature of the binomial model which causes spikes in density as the stock price can only take specific values. The stock price from the Black-Scholes model follows a continuous (log-normal) distribution, explaining its smoothness in comparison. Looking at the CDF, in figure 3, this supports this explanation further. Overall, this suggests the stock price distribution from the binomial model is a good fit to the true stock price distribution.

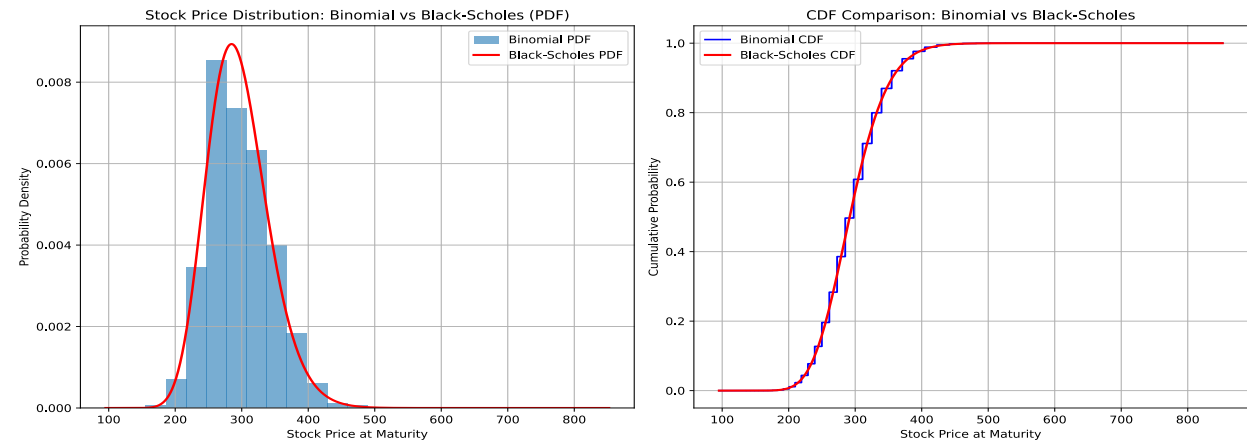


Figure 3: Stock Price Distribution from the Approximate Binomial CRR Model Compared to the Black-Scholes Model

We next explore the 'optimal' value of the up factor (u) - the value that minimises the absolute option price error. We start to do so by altering our original optimised function for the approximate binomial CRR model to take u as an argument. Then we test different u 's over a range of values and calculate option prices for the corresponding u and compare this price to that of the Black-Scholes model (the true value). We set the lower bound of this range of values to be $e^{r\Delta t} \times (1 + 10^{-6})$ to ensure the no arbitrage condition of $u > e^{r\Delta t}$ is met. We use this rather than simply $e^{r\Delta t}$ to ensure our condition of a strict inequality is met. We use an upper bound of 1.04. The point where the option price from our model equals the Black-Scholes price is where the 'optimal' u lies. Figure 4, illustrates this process in a plot for both a call and put option under our given parameters from earlier. We can see the 'optimal' u is 1.0221 in this case.

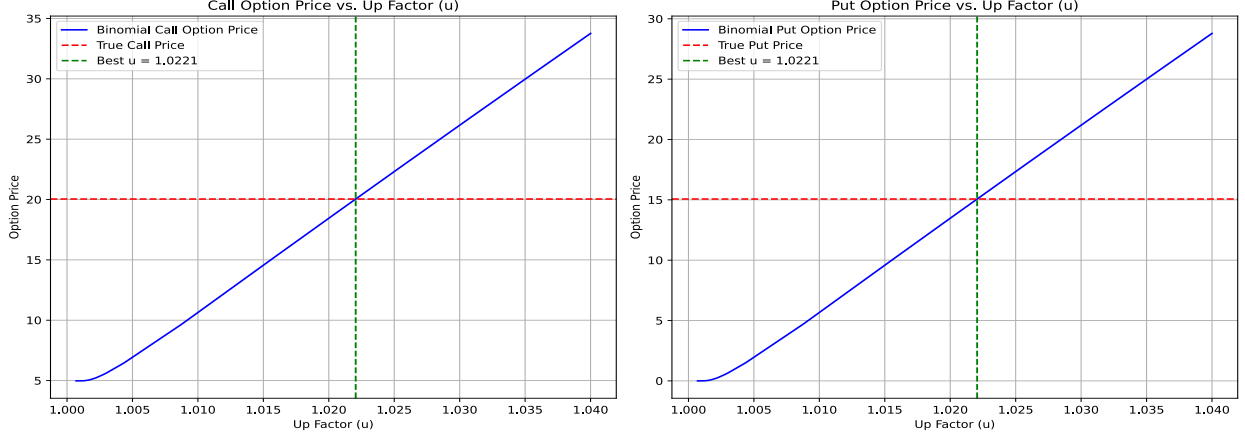


Figure 4: Option Price vs. Up Factor (u)

In table 4, we can see a comparison of the mean and second moment of the stock price from our model using the 'optimal' u and the Black-Scholes model. Similar to our original optimised function we have the same mean stock price to 4 decimal places and the second moment of the stock price is fairly close, this time with a greater absolute error at 22.2898. This makes theoretical sense, as the first moment matches because risk-neutral pricing ensures that the expected stock price remains consistent between the models. However, the second moment does not necessarily match, as selecting u to minimise absolute option price error does not explicitly enforce matching of the second moments, leading to small deviations from the true value.

Model	Mean Stock Price	Second Moment of Stock Price
Black-Scholes	295.1516	89230.5154
'Optimal' u	295.1516	89208.2256

Table 4: Comparison of Mean Stock Price and Second Moment of Stock Price for the 'optimal' u Function to the True Values

Overall, in practice, this approach for selection of the 'optimal' u factor for the binomial model isn't very useful, as it assumes we know the exact price (the Black-Scholes price). If we know the exact price there is no need for the implementation of an approximation like the binomial model, thus, this approach is rendered useless. It can also be noted that it is significantly more computationally expensive than the other approaches (highlighted by the computational times seen in table 5 and table 6) without any significant gain in accuracy.

Tilted Tree and Richardson Extrapolation

In this section, we adapt our original optimised function to use the exact CRR method, a tilted tree and Richardson extrapolation. Implementation of the exact CRR method involves a change of u and d in our original function to:

$$u = \beta + \sqrt{\beta^2 - 1}, \quad d = \frac{1}{u} = \beta - \sqrt{\beta^2 - 1}. \quad (7)$$

$$\text{where } \beta = \frac{1}{2} \left(e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t} \right),$$

Theory implies the first two moments of the stock price calculated from the exact CRR model must equal those from the Black-Scholes model as u and d are selected for this to be the case. From table 5 and table 6, we can see this holds in our application suggesting correct implementation of the exact CRR model.

We further verify the correct implementation of the exact binomial CRR model by plotting its convergence to the Black-Scholes price as M increases. As seen in figure 5, an irregular convergence to the Black-Scholes price occurs, which is expected from this model.

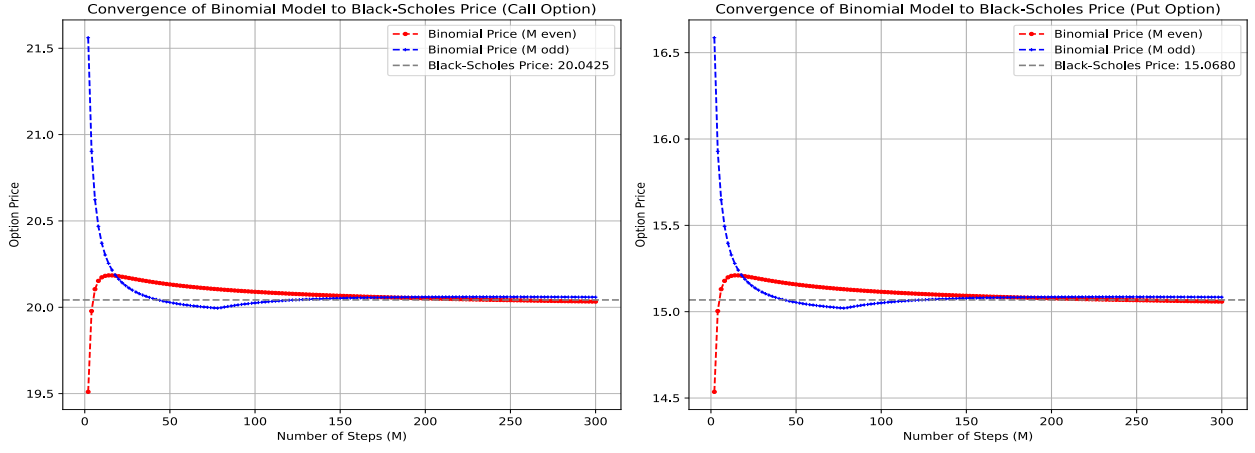


Figure 5: Convergence of Exact Binomial CRR to Black-Scholes Price

Tilted trees have the aim of improving the accuracy of the algorithm by increasing the speed the option price computed by the binomial model converges to the Black-Scholes price. As seen with the approximate model, convergence was irregular, so tilted trees aim to smoothen convergence and speed it up. To remove this irregularity in convergence we want the strike price of the stock (K) to fall exactly in the middle of the binomial tree. This is easy to achieve in the approximate CRR model, as u and d are chosen such that $ud = 1$, so we simply alter our original function to set u and d as:

$$u = e^{\sigma\sqrt{\Delta t} + \frac{1}{M}\log(K/S_0)}, \quad d = e^{-\sigma\sqrt{\Delta t} + \frac{1}{M}\log(K/S_0)}. \quad (8)$$

Figure 5, shows how the option price changes as M is increased, from which we can see a smooth convergence to the Black-Scholes price. This suggests our implementation of tilted trees is successful, as it improves upon the irregular convergence of the approximate CRR model that we saw in figure 2.

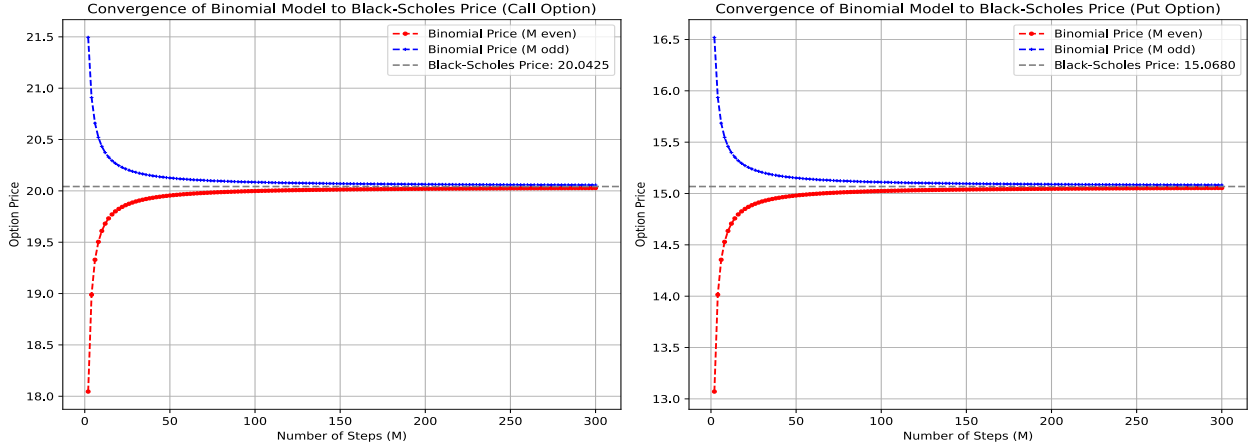


Figure 6: Convergence of Tilted Tree to Black-Scholes Price

Tilted trees are expected to improve convergence, however, computed prices aren't necessarily more accurate than the approximate binomial CRR method. Richardson extrapolation aims to improved the accuracy of the computed prices from the tilted tree method. It uses the idea that the computed price converges to the exact price as M increases, as can be seen from the following equation:

$$P(M) = P_{\text{ex}} + AM^{-1} + \mathcal{O}(M^{-2}). \quad (9)$$

This can be reformulated to the following:

$$2P(M) - P(M/2) = P_{\text{ex}} + \mathcal{O}(M^{-2}). \quad (10)$$

From this we can see we obtain a more accurate estimate of the exact price due to the loss of the leading-order error term AM^{-1} . Implementation of this method obviously increases computational time due to the need to run our tilted tree function twice (for M and $M/2$), but the increase in accuracy is expected to make up for this.

It can be noted that Richardson extrapolation only works in the case where M and $M/2$ are even because the leading-order error term AM^{-1} must be identical for both cases to ensure proper cancellation in equation (10). Figure 7, highlights this through the jumps closer and further from the Black-Scholes price when M is even. If M is odd cancellation is impossible, as $M/2$ will not be an integer. We tackle this in Python using floor division ($M // 2$). However, this isn't theoretically valid as it is a work around and we aren't truly using $M/2$ in our calculation. We see the same jumps for odd M in figure 7, as floor division causes misalignment of the error terms when M is odd and $M/2$ is even. Overall, even M and $M/2$ is the only case where implementation of Richardson extrapolation is theoretically valid, as otherwise there is misalignment of the leading-order error terms.

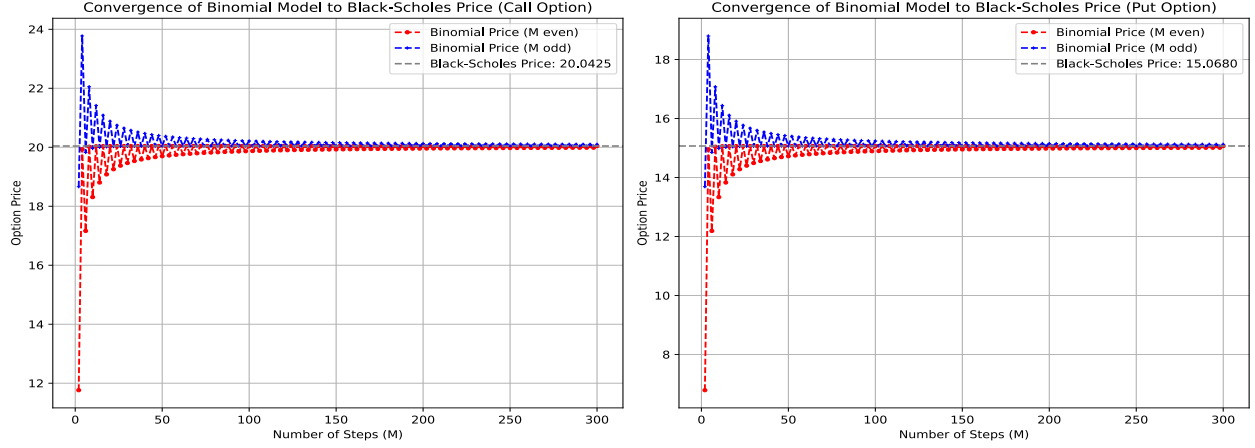


Figure 7: Convergence of Tilted Tree with Richardson Extrapolation to Black-Scholes Price

This explanation can be supported further by comparison of our results in table 5 and table 6. As can be seen in table 5, where M and $M/2$ are even (as $M = 48$), the absolute error of the option price (0.0005) is significantly lower than seen in table 6 (0.3462), where M is even and $M/2$ is odd (as $M = 50$).

Model	Option Price	Absolute Error	Mean Stock Price	Second Moment of Stock Price	Computational Time (s)
Black-Scholes	20.0425	0.0000	295.1516	89230.5154	0.0001
Approximate Binomial CRR	20.1176	0.0751	295.1516	89226.2348	0.0001
Exact Binomial CRR	20.1349	0.0924	295.1516	89230.5154	0.0001
Optimal u	20.0432	0.0007	295.1516	89207.9029	1.1675
Tilted Tree	19.9513	0.0912	295.1516	89228.7084	0.0001
Tilted Tree with Richardson Extrapolation	20.0420	0.0005	295.1516	89230.5131	0.0003

Table 5: Comparison of Option Pricing Models for a Call option($M=48$)

From table 5 we can see that in the case where both M and $M/2$ are even, tilted trees with Richardson extrapolation give the smallest absolute error. The computational time is three times that of the approximate binomial CRR, exact binomial CRR and tilted trees; however, the significant gain in accuracy makes up for this. We note for all models that the mean stock price is equal to the true value (due to risk-neutral pricing in all models). The second moment of the stock prices is slightly different, but close to the true value in all cases, except for the exact CRR model where it is the exact same (as u and d are chosen for it to be so). The tilted trees with Richardson extrapolation has the closest second moment to the true value, after the exact CRR model. Overall, the suggestion from this is that tilted trees with Richardson extrapolation is the best model for approximating the price of European options when considering all the factors mentioned.

Model	Option Price	Absolute Error	Mean Stock Price	Second Moment of Stock Price	Computational Time (s)
Black-Scholes	20.0425	0.0000	295.1516	89230.5154	0.0002
Approximate Binomial CRR	20.1157	0.0732	295.1516	89226.4058	0.0001
Exact Binomial CRR	20.1324	0.0899	295.1516	89230.5154	0.0001
Optimal u	20.0419	0.0005	295.1516	89208.2256	1.2426
Tilted Tree	19.9549	0.0875	295.1516	89228.7806	0.0001
Tilted Tree with Richardson Extrapolation	19.6962	0.3462	295.1516	89230.5132	0.0003

Table 6: Comparison of Option Pricing Models for a Call Option ($M=50$)

Table 6 offers a slightly different conclusion, due to the fact that M and $M/2$ are not both even. As explained, this leads to poor accuracy for tilted trees with Richardson extrapolation. In this case, the 'optimal' u function offers the smallest absolute error. However, as mentioned this is not practically viable, as it assumes knowledge of the true price for its calculation, rendering it useless. We can also note that it is substantially more computationally expensive than any of the other functions (12,426 times more than the approximate binomial CRR method). The behaviour of the mean and

second moment of the stock price across the models is the same as described for table 5. Overall, the suggestion from this is that the approximate binomial CRR model is the best model for approximating the price of European options in the case where M and $M/2$ are not both even, due to its practical viability and small absolute error.

Results and Discussion

The first conclusion we are able to draw is that when starting from the approximate binomial CRR model we can significantly improve computational complexity by taking advantage of the assumed binomial distribution of stock prices to build a function that uses no arrays. This theoretical ideology is supported by the practical implementation across a large range of M values. We then go on to verify the accuracy of our implementation of this methodology by implementing a function for calculation of the true price (from the Black-Scholes model) for comparison. We see the prices for both call and put options are fairly close to the true value under the given parameters ($S_0 = \text{£}285, r = 2.1\%, \sigma = 12\%, T = 20, K = \text{£}290, M = 50$). We strengthen these findings through plotting the convergence to the true price as M increases, in which we see an irregular convergence (as expected). We also compare the stock price distribution from the approximate binomial CRR model to the true stock price distribution (from the Black-Scholes model), and find it to be a good fit. We go on to investigate other models with the aim of improving accuracy and obtaining regular convergence to the true price.

We start by finding an 'optimal' u that minimises the absolute option price error. This does offer success in increasing accuracy but at the cost of a significant increase in computational time, as well as the lack of practical viability due to assumption of the knowledge of the true price. We next implement tilted trees with the aim of improving convergence to the true price, which we are able to achieve. In the attempt of improving upon the accuracy of tilted trees we implement Richardson extrapolation. We find that in the case where M and $M/2$ are both even we are able to do so, but in other cases this is not possible. In the successful case, the computational time is slightly more, however, the significant gain in accuracy makes up for this.

Overall, we conclude that in the case where M and $M/2$ are both even tilted trees with Richardson extrapolation offer the greatest combination between accuracy and computational complexity. In alternative cases, where M and $M/2$ are not both even, our initial approximate binomial CRR model offers the greatest combination between accuracy and computational complexity. It must be noted that these conclusions are in the cases where $M = 48$ and $M = 50$ under our given parameters. It is possible that if M is increased further tilted trees will offer the greatest combination between accuracy and computational complexity, in the case where M and $M/2$ are not both even, due to its smooth convergence to the true price, as opposed to the irregular convergence of the approximate binomial CRR model.